# CELUMUSA VINCENT MKHATSHWA



**INFORMATION TECHNOLOGY | DATA & BUSINESS INTELLIGENCE | EDUCATOR**

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Portfolio | GitHub | LinkedIn

## PROFESSIONAL PROFILE

Final-year BSc Information Technology student at North-West University and Mathematics educator with 7+ years of professional experience. Practical experience spanning full-stack web applications, SQL databases, Power BI-supported business analysis, Excel inventory optimisation, machine learning and Linux systems. Combines analytical problem-solving with strong communication, documentation, teamwork and leadership; interested in software, data analytics, business intelligence, information systems and technology-enabled business operations.

## TECHNICAL AND PROFESSIONAL SKILLS

**Programming & web:** Java, C#, Python, C++, JavaScript, SQL, HTML/CSS

**Software & UI:** React, Vite, Node.js, Express, ASP.NET, responsive design

**Data & BI:** Power BI visual analysis, Excel, Excel QM, statistics, decision support

**Databases:** Oracle SQL, SQL Server, MySQL, SQLite, relational design, joins, views, indexes

**AI, Systems & Security:** scikit-learn, K-Means, Teachable Machine, Ubuntu Linux, VirtualBox, Information Security.

**Delivery & tools:** Git/GitHub, Mapbox, Visual Studio, testing, technical documentation

## EDUCATION AND ACADEMIC ACHIEVEMENTS

**Bachelor of Science in Information Technology** | North-West University | Expected 2026

NQF 7 • 328/392 credits completed • 78.70% cumulative average • 22 distinctions (2024–2026) • 2026 average: 82.82%

**Bachelor of Education** | University of the Witwatersrand | 2018

Majors: Mathematics, Physical Sciences and Natural Sciences

## SELECTED PROJECTS

**PukParking - Smart Campus Parking System** | React • JavaScript • Node.js • Express • Mapbox • Git

Contributed to a campus parking web application; developed Home, About and Parking Statistics interfaces, improved navigation, responsive layouts and dark mode, supported testing/integration, and prepared user and technical manuals.

**Uber/Bolt South Africa - Mobility Intelligence** | Oracle SQL • Power BI • Data analysis

Collaborated on a relational database and analysis of 50,000 ride-hailing trips; interpreted Power BI visualisations and examined platform, city, cancellation and surge-pricing patterns, including a 33.34% overall cancellation rate.

**MPBC Bicycle Inventory Optimisation** | Excel • Excel QM • Decision support

Applied inventory models to estimate an economic order quantity of approximately 68 bicycles, a reorder point of 72 and safety stock of 38, supporting evidence-based stock control.

**Artificial Intelligence and Machine Learning Practicals** | Python • scikit-learn • Teachable Machine

Implemented K-Means clustering with scaling, elbow and silhouette evaluation; trained and evaluated image-based emotion classification models.

**Municipal Complaint Management System** | C# • ASP.NET • SQL Server

Contributed to a complaint submission and tracking system with administrative case allocation, service management and filtered reports.

**CMA Fitness Management Database** | SQL • Relational design

Led database modelling for memberships, bookings, trainers, payments, attendance and maintenance, using normalisation, tables, views, indexes and queries.

**Burns Beach Physio Management System** | C# • Web & Windows applications

Designed user-facing and administrative workflows for registration, appointments, referrals, invoices and reporting.

**Linux Operating Systems Practical** | Ubuntu • VirtualBox • Bash

Configured an Ubuntu virtual machine, Guest Additions and shared folders; demonstrated command-line utilities and documented practical results.

**Professional Virtual CV** | HTML • CSS • JavaScript • GitHub Pages

Developed a responsive personal portfolio with dark mode, accessible navigation, project highlights and a downloadable CV.

**Programming and Data Structures** | Java • C++ • Algorithms • OOP

Built practice applications using inheritance, interfaces, custom exceptions, lists, stacks, queues and file handling; applied core data-structure concepts.

## PROFESSIONAL EXPERIENCE

**Mathematics Educator | Rembrandt Park Primary School** | 2019 - Present

- **Achievement:** Maintained a 100% Mathematics pass rate with class averages above 70%.
- **Teaching & technology:** Deliver Mathematics instruction, support learners, and integrate digital tools plus coding and robotics activities.
- **Coordination:** Manage assessments, records and communication with parents and school leadership; support sport and district-level activities.

**Temporary Mathematics Educator | Hillview High School** | Jan - Feb 2019

Taught secondary-school Mathematics and supported foundational and problem-solving skills.

## LEADERSHIP AND CERTIFICATIONS

**Leadership:** District Area Manager, Johannesburg Primary Schools Football Association (JPSFA); sports coaching, fixture coordination and stakeholder communication.

**IT Certifications:**

- Introduction to Cloud Security - Simplilearn SkillUp (May 2025)
- Introduction to Cyber Security - Simplilearn SkillUp (May 2025)

**Other Certifications:** SAFA Level 2 Referee (2025); Project-Based Learning (2025); Cubroid Coding & Robotics (2024); Positive Behaviour Management (2024); JPSFA Coaching (2023); 120-Hour TEFL, Wits (2020).

## PROFESSIONAL REFERENCES

**Virna Slaughter**
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